UNIT 4 · ARS/NET PATTERN

MCQ Bank — 210 Questions with Answers

Thermodynamics, power cycles and fuels (Q1–Q35)

Q1. For air, the ratio of specific heats γ is:

(a) 1.2 (b) 1.3
(c) 1.4 (d) 1.67

Q2. cp - cv equals:

(a) γ (b) R
(c) U (d) H

Q3. In an isochoric process, work done is:

(a) pΔV (b) Zero
(c) Maximum (d) Negative always

Q4. Ideal engine compression/expansion is modelled as:

(a) Isothermal (b) Isobaric
(c) Isentropic (pV^γ = const) (d) Isochoric

Q5. The first law of thermodynamics states:

(a) Entropy rises (b) Q = ΔU + W (energy conservation)
(c) Heat flows cold→hot (d) Work is heat

Q6. Maximum possible efficiency between 900 K and 300 K is:

(a) 33.3% (b) 50%
(c) 66.7% (d) 100%

Q7. Heat addition in the Otto cycle occurs at constant:

(a) Pressure (b) Volume
(c) Temperature (d) Entropy

Q8. Heat addition in the Diesel cycle occurs at constant:

(a) Volume (b) Pressure
(c) Entropy (d) Enthalpy

Q9. Air-standard Otto efficiency is:

(a) 1 - r^(γ-1) (b) 1 - 1/r^(γ-1)
(c) 1/r^γ (d) 1 - T2/T1 always

Q10. For r = 8 and γ = 1.4, Otto efficiency ≈

(a) 46.5% (b) 51%
(c) 56.5% (d) 62%

Q11. Raising the cut-off ratio ρ (higher load) makes Diesel-cycle efficiency:

(a) Rise (b) Fall
(c) Constant (d) Double

Q12. For the SAME compression ratio, the most efficient cycle is:

(a) Diesel (b) Dual
(c) Otto (d) All equal

Q13. For the same MAXIMUM pressure and temperature, the most efficient is:

(a) Otto
(b) Diesel
(c) Dual
(d) Equal

Q14. Modern high-speed diesels are best represented by the:

(a) Otto cycle
(b) Dual (limited-pressure) cycle
(c) Carnot
(d) Brayton

Q15. SI-engine compression ratio is limited by:

(a) Cost
(b) Knocking of end gas
(c) Weight
(d) Cetane

Q16. Typical CI (diesel) compression ratios are:

(a) 4–6
(b) 6–10
(c) 14–22
(d) 25–35

Q17. Diesels can use high r because:

(a) Fuel is compressed
(b) Only AIR is compressed — fuel injected later
(c) They run cold
(d) They are heavy

Q18. Mean effective pressure equals:

(a) Peak pressure
(b) Net cycle work ÷ swept volume
(c) Injection pressure
(d) Boost

Q19. Cetane number measures:

(a) Knock resistance of petrol
(b) Ignition quality (short delay) of diesel
(c) Volatility
(d) Viscosity

Q20. Minimum cetane number of BS-VI Indian HSD is:

(a) 40
(b) 45
(c) 51
(d) 60

Q21. BS-VI fuel sulphur limit is:

(a) 10 ppm
(b) 50 ppm
(c) 350 ppm
(d) 500 ppm

Q22. Calorific value of diesel is about:

(a) 27
(b) 35
(c) 42–45
(d) 55 MJ/kg

Q23. A fuel with HIGH octane number has cetane number that is:

(a) High
(b) Low
(c) Same
(d) Unrelated

Q24. Octane rating scale uses iso-octane = 100 and:

(a) Benzene = 0
(b) n-heptane = 0
(c) Cetane = 0
(d) Methane = 0

Q25. Diesel engines control load by:

(a) Throttling air
(b) Varying fuel injected into unthrottled air (lean overall)
(c) Spark timing
(d) Valve lift only

Q26. 'Diesel knock' results from:

(a) Too short ignition delay
(b) Too long ignition delay (low cetane)
(c) High octane
(d) Rich mixture

Q27. Calorific value of fuels is measured with a:

(a) Patternator (b) Bomb calorimeter
(c) Dynamometer (d) Hydrometer

Q28. Hydrogen's calorific value per kg is roughly:

(a) Equal to diesel (b) Half of diesel
(c) Nearly 3× diesel (~120 MJ/kg) (d) 10× diesel

Q29. Biodiesel is produced from vegetable oils by:

(a) Distillation (b) Transesterification
(c) Cracking only (d) Fermentation

Q30. Ethanol blending of 20% (E20) became mandatory across India from:

(a) April 2020 (b) April 2023
(c) April 2026 (d) 2030

Q31. India achieved the 20% average ethanol-blending milestone in:

(a) 2020 (b) 2023
(c) November 2025 (d) Not yet

Q32. TREM-IV norms (Jan 2023) applied to tractors of:

(a) All sizes (b) Above 50 hp only
(c) Below 20 hp (d) Combines only

Q33. TREM-V norms for 56–560 kW tractors/combines take effect from:

(a) Jan 2023 (b) 1 Oct 2026
(c) 2030 (d) 2035

Q34. Under TREM-V the PM limit for major categories tightens to:

(a) 0.15 (b) 0.075
(c) 0.025 (d) 0.015 g/kWh

Q35. TREM-V aligns Indian tractor emissions with:

(a) US Tier 2 (b) EU Stage V
(c) BS-III (d) China-IV

Answers

Q1 — (c) 1.4
Q2 — (b) R
Q3 — (b) Zero
Q4 — (c) Isentropic (pV^γ = const)
Q5 — (b) Q = ΔU + W (energy conservation)
Q6 — (c) 66.7% — *η = 1 - 300/900 = 2/3 = 66.7%.*
Q7 — (b) Volume
Q8 — (b) Pressure
Q9 — (b) 1 - 1/r^(γ-1)
Q10 — (c) 56.5% — *8^0.4 = 2.297 → 1 - 0.435 = 56.5%.*
Q11 — (b) Fall
Q12 — (c) Otto — *At equal r, constant-volume (Otto) heat addition is thermodynamically superior.*
Q13 — (b) Diesel — *At equal peak pressure, the Diesel cycle rejects less heat — the classic viva pair with Q12.*
Q14 — (b) Dual (limited-pressure) cycle
Q15 — (b) Knocking of end gas
Q16 — (c) 14–22
Q17 — (b) Only AIR is compressed — fuel injected later

Q18 — (b) Net cycle work ÷ swept volume

Q19 — (b) Ignition quality (short delay) of diesel

Q20 — (c) 51

Q21 — (a) 10 ppm

Q22 — (c) 42–45

Q23 — (b) Low

Q24 — (b) n-heptane = 0

Q25 — (b) Varying fuel injected into unthrottled air (lean overall)

Q26 — (b) Too long ignition delay (low cetane) — *A long delay lets too much fuel accumulate → sudden premixed burn → pressure shock; cured by higher cetane, a warm engine, higher r.*

Q27 — (b) Bomb calorimeter

Q28 — (c) Nearly 3× diesel (~120 MJ/kg)

Q29 — (b) Transesterification

Q30 — (c) April 2026 — *From the updated information box (PIB/MoPNG; MoRTH TREM notifications; verified July 2026).*

Q31 — (c) November 2025

Q32 — (b) Above 50 hp only

Q33 — (b) 1 Oct 2026 — *As given in the source notes (verified July 2026) — watch for further TREM-V deferral news for the 25–50 hp class.*

Q34 — (d) 0.015 g/kWh

Q35 — (b) EU Stage V

IC engine systems — operation, adjustment and troubleshooting (Q36–Q70)

Q36. In a 4-stroke engine the camshaft rotates at:

(a) Crank speed (b) Half crank speed

(c) Twice crank speed (d) Quarter speed

Q37. A 4-stroke engine gives one power stroke per:

(a) 1 revolution (b) 2 revolutions

(c) 4 revolutions (d) ½ revolution

Q38. Bore 100 mm, stroke 120 mm → swept volume per cylinder ≈

(a) 0.55 L (b) 0.79 L

(c) 0.94 L (d) 1.2 L

Q39. With Vc = 55 cm³ and Vs = 942 cm³, compression ratio ≈

(a) 14 (b) 16

(c) 18 (d) 22

Q40. Valve overlap is the period when:

(a) Both valves are closed (b) Both intake and exhaust are open near TDC

(c) Piston stops (d) Injection occurs

Q41. The exhaust valve opens:

(a) At BDC exactly (b) 30–50° BEFORE BDC

(c) At TDC (d) After TDC only

Q42. Too SMALL a valve (tappet) clearance causes:

(a) Noise only (b) Valve leakage and burnt valves

(c) Higher power (d) Better idle

Q43. Too LARGE a clearance causes:

(a) Burnt valves
(b) Noisy operation with late opening and reduced breathing
(c) Overheating
(d) Oil dilution

Q44. Valve clearance is measured with a:

(a) Micrometer
(b) Feeler gauge
(c) Dial bore gauge
(d) Vernier

Q45. The decompression lever:

(a) Increases r
(b) Holds valves open to ease cranking
(c) Stops fuel
(d) Adjusts governor

Q46. In an inline FIP, fuel quantity is metered by:

(a) Throttle plate
(b) Rotating plunger helix via the control rack
(c) Venturi
(d) Float

Q47. Conventional diesel injector opening pressure is about:

(a) 20–50 bar
(b) 170–250 bar
(c) 1000–1500 bar
(d) 5 bar

Q48. After changing fuel filters, a diesel must be:

(a) Timed
(b) Bled of air (primed)
(c) Rebored
(d) Decoked

Q49. An injector is bench-tested for:

(a) Opening pressure, spray pattern, chatter, drip
(b) Colour
(c) Weight
(d) Torque

Q50. A choked air cleaner produces:

(a) White smoke
(b) Black smoke and power loss
(c) Blue smoke
(d) No effect

Q51. The oil-bath air cleaner is being replaced by:

(a) No cleaner
(b) Dry paper element with cyclonic pre-cleaner
(c) Water bath
(d) Sponge only

Q52. A turbocharger is driven by:

(a) Crankshaft belt
(b) Exhaust gases
(c) Camshaft
(d) PTO

Q53. Diesel engines use which method of governing?

(a) Quantity (throttle)
(b) Quality (vary fuel, air unthrottled)
(c) Hit and miss
(d) Spark control

Q54. Rhythmic surging of engine speed due to over-sensitive governor is:

(a) Droop
(b) Hunting
(c) Knocking
(d) Dieseling

Q55. Governor droop (regulation) is the:

(a) Oil leak
(b) Speed fall from no-load to full-load as %
(c) Belt slip
(d) Rack travel

Q56. The thermostat begins to open at about:

(a) 40 °C
(b) 60 °C
(c) 82 °C
(d) 100 °C

Q57. The radiator pressure cap:

(a) Cools the fan
(b) Raises coolant boiling point
(c) Filters coolant
(d) Drives the pump

Q58. Thermosiphon cooling works by:

(a) Pump pressure
(b) Density difference between hot and cold water
(c) Fan suction
(d) Thermostat

Q59. Overcooling an engine causes:

(a) Better economy
(b) Crankcase oil dilution, condensate corrosion and wear
(c) More power
(d) Nothing

Q60. Fan-belt tension is checked by mid-span deflection of about:

(a) 2–3 mm
(b) 12–20 mm
(c) 50 mm
(d) 100 mm

Q61. Tractor engines use which lubrication system?

(a) Petroil mixing
(b) Splash only
(c) Forced-feed full-pressure
(d) Gravity

Q62. Normal engine oil pressure at rated speed is about:

(a) 0.2–0.4
(b) 2–4
(c) 20–40
(d) 100 bar

Q63. 'SAE 15W-40' describes the oil's:

(a) API service class
(b) Viscosity grade (multigrade)
(c) Flash point
(d) Density

Q64. Blue exhaust smoke indicates:

(a) Excess fuel
(b) Burning lubricating oil (worn rings/guides)
(c) Water in fuel
(d) Retarded spark

Q65. White smoke from a diesel commonly indicates:

(a) Overload
(b) Misfire/unburnt fuel or water — low compression, cold start
(c) Oil burning
(d) Rich mixture

Q66. Black smoke under load indicates:

(a) Excess fuel/insufficient air — overload, choked air cleaner, bad injector
(b) Oil
(c) Coolant
(d) Normal

Q67. A magneto ignition system:

(a) Needs a battery
(b) Generates its own ignition current
(c) Works only on diesels
(d) Uses glow plugs

Q68. Spark plug gap is typically:

(a) 0.06–0.08 mm
(b) 0.6–0.8 mm
(c) 6–8 mm
(d) 2–3 mm

Q69. Cold-start aid in diesel engines:

(a) Choke
(b) Glow plugs / intake heater
(c) Carburettor
(d) Rich jet

Q70. Diesel knock is reduced by:

(a) Lower cetane fuel

(b) Higher cetane fuel, warm engine, higher compression temperature

(c) Bigger fan

(d) Lean mixture

Answers

Q36 — (b) Half crank speed

Q37 — (b) 2 revolutions

Q38 — (c) 0.94 L — *$Vs = (\pi/4)(1.0)^2(1.2)\ dm^3 = 0.942\ L$.*

Q39 — (c) 18 — *$r = (942 + 55)/55 = 18.1$ — a typical DI diesel.*

Q40 — (b) Both intake and exhaust are open near TDC

Q41 — (b) 30–50° BEFORE BDC

Q42 — (b) Valve leakage and burnt valves — *Clearance compensates thermal expansion; burnt valves (too small) vs noisy tappets (too large) is a perennial exam pair.*

Q43 — (b) Noisy operation with late opening and reduced breathing

Q44 — (b) Feeler gauge

Q45 — (b) Holds valves open to ease cranking

Q46 — (b) Rotating plunger helix via the control rack

Q47 — (b) 170–250 bar

Q48 — (b) Bled of air (primed)

Q49 — (a) Opening pressure, spray pattern, chatter, drip

Q50 — (b) Black smoke and power loss

Q51 — (b) Dry paper element with cyclonic pre-cleaner

Q52 — (b) Exhaust gases

Q53 — (b) Quality (vary fuel, air unthrottled)

Q54 — (b) Hunting

Q55 — (b) Speed fall from no-load to full-load as %

Q56 — (c) 82 °C

Q57 — (b) Raises coolant boiling point

Q58 — (b) Density difference between hot and cold water

Q59 — (b) Crankcase oil dilution, condensate corrosion and wear

Q60 — (b) 12–20 mm

Q61 — (c) Forced-feed full-pressure

Q62 — (b) 2–4

Q63 — (b) Viscosity grade (multigrade)

Q64 — (b) Burning lubricating oil (worn rings/guides) — *Smoke triage: BLUE = oil, BLACK = fuel-rich/air-starved, WHITE = unburnt/misfire/water.*

Q65 — (b) Misfire/unburnt fuel or water — low compression, cold start

Q66 — (a) Excess fuel/insufficient air — overload, choked air cleaner, bad injector

Q67 — (b) Generates its own ignition current

Q68 — (b) 0.6–0.8 mm

Q69 — (b) Glow plugs / intake heater

Q70 — (b) Higher cetane fuel, warm engine, higher compression temperature — *Higher cetane shortens the ignition delay, so less fuel accumulates before first ignition — the opposite of SI knock chemistry (Topic 1b).*

Engine calculations — power, torque, firing order, heat load, piston to

flywheel (Q71–Q105)

Q71. Torque (N·m) from power P (kW) at N rpm is:

(a) 9550·P/N
(b) P·N/9550
(c) 60·P/N
(d) 746·P/N

Q72. BP = 45 kW at 2200 rpm gives torque ≈

(a) 145
(b) 175
(c) 195
(d) 225 N·m

Q73. In the IP formula, n for a 4-stroke engine equals:

(a) N
(b) N/2
(c) 2N
(d) N/4

Q74. p_m = 700 kPa, L = 0.12 m, A = 78.54 cm², 4 cylinders, 2200 rpm → IP ≈

(a) 24 kW
(b) 36 kW
(c) 48 kW
(d) 60 kW

Q75. If BP = 40 kW for the engine of Q4, mechanical efficiency ≈

(a) 72%
(b) 78%
(c) 83%
(d) 90%

Q76. Mean piston speed equals:

(a) LN/60
(b) 2LN/60
(c) πLN/60
(d) 4LN/60

Q77. L = 120 mm at 2200 rpm → mean piston speed =

(a) 4.4
(b) 8.8
(c) 13.2
(d) 17.6 m/s

Q78. Torque backup of an engine indicates its:

(a) Fuel economy
(b) Lugging ability under overload
(c) Oil pressure
(d) Idle quality

Q79. T_{rated} = 195 N·m, T_{max} = 240 N·m → torque backup ≈

(a) 12%
(b) 18%
(c) 23%
(d) 45%

Q80. As load slows a diesel below rated speed, its torque:

(a) Falls
(b) Rises toward T_{max}
(c) Is constant
(d) Becomes zero

Q81. Firing interval of a 4-stroke, 4-cylinder engine is:

(a) 90°
(b) 120°
(c) 180°
(d) 240°

Q82. Firing interval of a 4-stroke, 3-cylinder engine is:

(a) 120°
(b) 180°
(c) 240°
(d) 360°

Q83. Firing interval of a 4-stroke, 6-cylinder engine is:

(a) 60°
(b) 90°
(c) 120°
(d) 180°

Q84. The most common firing order of 4-cylinder tractor engines is:

(a) 1-2-3-4
(b) 1-3-4-2
(c) 4-3-2-1
(d) 1-4-3-2

Q85. Standard firing order of an inline 6-cylinder engine is:

(a) 1-2-3-4-5-6 (b) 1-5-3-6-2-4

(c) 6-5-4-3-2-1 (d) 1-6-2-5-3-4

Q86. In a 4-cylinder engine, crank throws are paired:

(a) 1&2, 3&4 (b) 1&3, 2&4

(c) 1&4, 2&3 (d) Random

Q87. With order 1-3-4-2, when cylinder 1 is on POWER, cylinder 3 is on:

(a) Exhaust (b) Compression

(c) Suction (d) Power

Q88. And cylinder 2 is on:

(a) Exhaust (b) Compression

(c) Suction (d) Power

Q89. Firing orders are chosen mainly to:

(a) Save fuel (b) Distribute power impulses along the crank and balance bearing/manifold loads

(c) Reduce cost (d) Ease casting

Q90. In a 2-stroke, 3-cylinder engine, firing interval is:

(a) 240° (b) 180°

(c) 120° (d) 360°

Q91. At rated load, the useful (brake) share of fuel energy is roughly:

(a) 10% (b) 33%

(c) 60% (d) 90%

Q92. Heat carried away by cooling water is roughly:

(a) 5% (b) 15%

(c) 30% (d) 60% of fuel energy

Q93. Fuel 10 kg/h of CV 42,500 kJ/kg → energy input ≈

(a) 68 kW (b) 88 kW

(c) 118 kW (d) 425 kW

Q94. Coolant flow 1 kg/s with ΔT = 7 °C removes about:

(a) 7 kW (b) 15 kW

(c) 29 kW (d) 58 kW

Q95. Radiator heat load is roughly equal to:

(a) Twice IP (b) The engine's brake power

(c) Exhaust heat only (d) Zero

Q96. Side thrust of the piston on the cylinder wall equals:

(a) $F \cdot \sin\varphi$ (b) $F \cdot \tan\varphi$

(c) $F \cdot \cos\varphi$ (d) $F/\cos\varphi$

Q97. Piston side thrust is the main cause of:

(a) Valve wear (b) Piston slap and liner wear

(c) Injector choking (d) Radiator scale

Q98. Crankpin torque is produced by the:

(a) Radial force (b) Tangential component of connecting-rod thrust × crank radius

(c) Gas force alone (d) Cam force

Q99. Torque from a single cylinder is ZERO at:

(a) Mid stroke (b) Dead centres
(c) 30° ATDC (d) All angles

Q100. The connecting rod-to-crank ratio n = l/r is typically:

(a) 1–2 (b) 3.5–4.5
(c) 8–10 (d) 15

Q101. Secondary inertia force has frequency:

(a) Half crank speed (b) Crank speed
(c) Twice crank speed (d) Four times

Q102. Flywheel size is LARGEST for:

(a) 6-cylinder (b) 4-cylinder
(c) Single-cylinder engines (d) All equal

Q103. Required flywheel inertia is I =

(a) $\Delta E \cdot C_s \cdot \omega^2$ (b) $\Delta E/(C_s \cdot \omega^2)$
(c) $C_s \cdot \omega^2/\Delta E$ (d) $\Delta E \cdot \omega / C_s$

Q104. ΔE = 2000 J, C_s = 0.02, ω = 230 rad/s → I ≈

(a) 0.9 (b) 1.9
(c) 3.8 (d) 9.5 kg·m²

Q105. The turning-moment diagram of a multi-cylinder engine compared with a single is:

(a) More fluctuating (b) Smoother with overlapping impulses
(c) Identical (d) Flat zero

Answers

Q71 — (a) 9550·P/N
Q72 — (c) 195 — *T = 9550 × 45/2200 = 195.3 N·m.*
Q73 — (b) N/2
Q74 — (c) 48 kW — *0.66 kJ/stroke × 1100/60 × 4 = 48.4 kW.*
Q75 — (c) 83% — *40/48.4 = 82.6%.*
Q76 — (b) 2LN/60
Q77 — (b) 8.8 — *2 × 0.12 × 2200/60 = 8.8 m/s.*
Q78 — (b) Lugging ability under overload
Q79 — (c) 23% — *45/195 = 23%.*
Q80 — (b) Rises toward T_max
Q81 — (c) 180°
Q82 — (c) 240°
Q83 — (c) 120°
Q84 — (b) 1-3-4-2
Q85 — (b) 1-5-3-6-2-4
Q86 — (c) 1&4, 2&3
Q87 — (b) Compression — *Order 1-3-4-2 at 180° spacing: cylinder 3 fires next, so it is compressing now.*
Q88 — (a) Exhaust — *Cylinder 2 is last in the order — it fired 180° before cylinder 1, so it is now on exhaust (see the stroke chart).*
Q89 — (b) Distribute power impulses along the crank and balance bearing/manifold loads
Q90 — (c) 120°
Q91 — (b) 33%
Q92 — (c) 30%
Q93 — (c) 118 kW — *10 × 42,500/3600 = 118 kW.*

Q94 — (c) 29 kW — *1 × 4.187 × 7 = 29.3 kW.*
Q95 — (b) The engine's brake power
Q96 — (b) F·tanφ
Q97 — (b) Piston slap and liner wear
Q98 — (b) Tangential component of connecting-rod thrust × crank radius
Q99 — (b) Dead centres
Q100 — (b) 3.5–4.5
Q101 — (c) Twice crank speed
Q102 — (c) Single-cylinder engines
Q103 — (b) ΔE/(C_s·ω²)
Q104 — (b) 1.9 — *2000/(0.02 × 230²) = 1.89 kg·m².*
Q105 — (b) Smoother with overlapping impulses

Tractor power transmission, differential, final drives, PTO and drawbar (Q106–Q140)

Q106. The correct tractor power flow is:

(a) Engine→gearbox→clutch→diff
(b) Engine→clutch→gearbox→differential→final drive→wheels
(c) Engine→diff→clutch
(d) Engine→final drive→gearbox

Q107. Overall tractor transmission efficiency is about:

(a) 60–70%
(b) 75–80%
(c) 88–92%
(d) 100%

Q108. Torque capacity of a plate clutch is:

(a) μWR_m
(b) n·μ·W·R_m
(c) W/μR
(d) nWR/μ

Q109. For a single-plate clutch with both faces active, n =

(a) 1
(b) 2
(c) 4
(d) 0.5

Q110. μ = 0.3, W = 3000 N, Ro = 150 mm, Ri = 100 mm, single plate: T ≈

(a) 112
(b) 168
(c) 225
(d) 450 N·m

Q111. Clutch capacity is designed at about:

(a) 0.8×
(b) 1.0×
(c) 1.25–1.5×
(d) 3× engine torque

Q112. Free pedal play of a tractor clutch is about:

(a) 2–5 mm
(b) 20–25 mm
(c) 60–80 mm
(d) 100 mm

Q113. Insufficient free play causes:

(a) Hard steering
(b) Clutch slip and burnt friction plate
(c) More power
(d) Diff lock

Q114. A DUAL clutch on tractors provides:

(a) Four-wheel drive
(b) Live PTO (ground drive and PTO released in two stages)
(c) Better brakes
(d) Higher speed

Q115. Gear speeds are matched by friction cones in a:

(a) Sliding-mesh box
(b) Synchromesh gearbox
(c) Chain drive
(d) Belt CVT

Q116. In a constant-mesh gearbox, ratios are selected by:

(a) Sliding spur gears
(b) Dog clutches
(c) Belts
(d) Hydraulic rams

Q117. A creeper gear provides speeds of about:

(a) <1 km/h
(b) 5 km/h
(c) 15 km/h
(d) 30 km/h

Q118. Engine 2200 rpm, total reduction 60:1 → wheel speed ≈

(a) 18.3
(b) 36.7
(c) 55
(d) 73 rpm

Q119. With 1.35-m wheels at 36.7 rpm, travel speed ≈

(a) 4.7
(b) 9.3
(c) 14
(d) 18.6 km/h

Q120. Wheel torque = engine torque ×

(a) Reduction only
(b) Reduction × transmission efficiency
(c) Efficiency only
(d) Speed

Q121. The crown wheel and bevel pinion also provide:

(a) Straight drive
(b) Right-angle drive with large reduction
(c) PTO drive
(d) Steering

Q122. The differential speed relation is:

(a) $N_L - N_R = N_{crown}$
(b) $N_L + N_R = 2N_{crown}$
(c) $N_L \times N_R = N^2$
(d) $N_L = N_R$ always

Q123. Crown wheel 100 rpm and inner wheel 85 rpm → outer wheel:

(a) 100
(b) 105
(c) 115
(d) 130 rpm

Q124. A standard differential splits TORQUE between half axles:

(a) By speed
(b) Equally
(c) To the faster wheel
(d) To the heavier wheel

Q125. When one drive wheel spins in mud, the differential:

(a) Sends full torque to the gripping wheel
(b) Limits both wheels to the slipping wheel's torque
(c) Locks itself
(d) Stops the engine

Q126. The differential lock:

(a) Stops the tractor
(b) Couples both half-axles to rotate together
(c) Frees the PTO
(d) Applies brakes

Q127. Before turning at headland, the diff lock must be:

(a) Engaged
(b) Disengaged
(c) Oiled
(d) Doubled

Q128. The final drive's purposes include:

(a) Final large reduction near wheels and reduced load on diff shafts
(b) Cooling
(c) Fuel metering
(d) Suspension

Q129. Planetary final drives are located:

(a) In the gearbox
(b) In the wheel hubs
(c) On the PTO
(d) At the radiator

Q130. Tractor steering brakes act on the:

(a) Front wheels
(b) Bull-pinion/differential shafts of each rear wheel
(c) PTO
(d) Crankshaft

Q131. For road transport, brake pedals must be:

(a) Independent
(b) Latched (locked) together
(c) Removed
(d) Disconnected

Q132. Standard PTO speeds are:

(a) 540 and 1000 rpm
(b) 750 and 1500
(c) 300 and 600
(d) 1000 and 2000

Q133. The 540-rpm PTO shaft has:

(a) 21 splines, 45 mm
(b) 6 splines, 35 mm
(c) 8 splines, 25 mm
(d) Keyed 40 mm

Q134. The 1000-rpm PTO shaft has:

(a) 6 splines
(b) 10 splines
(c) 21 splines
(d) No splines

Q135. A PTO that can be engaged/disengaged anytime via its own clutch is:

(a) Transmission type
(b) Live
(c) Independent
(d) Ground-speed

Q136. With a TRANSMISSION-type PTO, pressing the clutch:

(a) Stops only wheels
(b) Stops both wheels and PTO
(c) Stops only PTO
(d) Nothing

Q137. Ground-speed PTO rotation is proportional to:

(a) Engine speed
(b) Forward travel of the tractor
(c) Fan speed
(d) Oil pressure

Q138. PTO power as a share of engine power is roughly:

(a) 50–60%
(b) 70–75%
(c) 87–92%
(d) 100%

Q139. Tractor drawbar height is typically:

(a) 100–150 mm
(b) 330–500 mm
(c) 700–900 mm
(d) 1200 mm

Q140. A swinging drawbar is used because it:

(a) Adds power
(b) Aligns with the line of pull of trailed implements on turns
(c) Cools the diff
(d) Raises speed

Answers

Q106 — (b) Engine→clutch→gearbox→differential→final drive→wheels

Q107 — (c) 88–92%

Q108 — (b) $n \cdot \mu \cdot W \cdot R_m$

Q109 — (b) 2

Q110 — (c) 225 — *$R_m = (0.15 + 0.10)/2 = 0.125$ m; $T = 2 \times 0.3 \times 3000 \times 0.125 = 225$ N·m.*

Q111 — (c) 1.25–1.5×

Q112 — (b) 20–25 mm
Q113 — (b) Clutch slip and burnt friction plate
Q114 — (b) Live PTO (ground drive and PTO released in two stages)
Q115 — (b) Synchromesh gearbox
Q116 — (b) Dog clutches
Q117 — (a) <1 km/h
Q118 — (b) 36.7 — *2200/60 = 36.7 rpm.*
Q119 — (b) 9.3 — *v = π × 1.35 × 36.7 = 155.6 m/min = 9.3 km/h.*
Q120 — (b) Reduction × transmission efficiency
Q121 — (b) Right-angle drive with large reduction
Q122 — (b) N_L+N_R = 2N_crown
Q123 — (c) 115 — *N_outer = 2 × 100 - 85 = 115 rpm.*
Q124 — (b) Equally
Q125 — (b) Limits both wheels to the slipping wheel's torque — *The spider gears balance torque — grip is limited by the WORSE wheel; hence the diff-lock pedal in wet fields.*
Q126 — (b) Couples both half-axles to rotate together
Q127 — (b) Disengaged
Q128 — (a) Final large reduction near wheels and reduced load on diff shafts
Q129 — (b) In the wheel hubs
Q130 — (b) Bull-pinion/differential shafts of each rear wheel
Q131 — (b) Latched (locked) together
Q132 — (a) 540 and 1000 rpm
Q133 — (b) 6 splines, 35 mm — *IS 4931/ISO 500 — 540 rpm: 35 mm, 6-spline; reached near 75% of rated engine speed.*
Q134 — (c) 21 splines — *IS 4931/ISO 500 — 1000 rpm: 21-spline.*
Q135 — (c) Independent
Q136 — (b) Stops both wheels and PTO
Q137 — (b) Forward travel of the tractor
Q138 — (c) 87–92%
Q139 — (b) 330–500 mm
Q140 — (b) Aligns with the line of pull of trailed implements on turns

Chassis mechanics, hitching, hydraulic position and draft control, steering (Q141–Q175)

Q141. For CG at distance x ahead of the rear axle and wheelbase L, front-axle reaction is:

(a) W·L/x
(b) W·x/L
(c) W(L-x)/L
(d) W/2

Q142. W = 20 kN, L = 2 m, x = 0.8 m → rear axle carries:

(a) 8 kN
(b) 10 kN
(c) 12 kN
(d) 16 kN

Q143. A 2WD tractor's static weight split front:rear is about:

(a) 50:50
(b) 65:35
(c) 35:65
(d) 20:80

Q144. Weight transferred rearward by pull P at drawbar height h is:

(a) P·L/h
(b) P·h/L
(c) P/(hL)
(d) W·h/L

Q145. P = 6 kN at h = 0.4 m, L = 2 m → weight transfer =

(a) 0.6
(b) 1.2
(c) 2.4
(d) 4.8 kN

Q146. Rearward overturning of a tractor is promoted by:

(a) Low drawbar hitching
(b) Hitching high above the drawbar and rear-axle torque reaction
(c) Front ballast
(d) Wide track

Q147. For steering control, the front axle should keep at least about:

(a) 5%
(b) 20%
(c) 50%
(d) 70% of tractor weight

Q148. Weight transfer during pulling is:

(a) Entirely harmful
(b) Useful for rear traction but dangerous in excess
(c) Irrelevant
(d) Zero at high pull

Q149. The three-point linkage consists of:

(a) 3 top links
(b) Two lower draft links + one top link
(c) One chain
(d) 4 arms

Q150. Category-I linkage mast height is:

(a) 360 mm
(b) 460 mm
(c) 610 mm
(d) 810 mm

Q151. Category-II lower hitch pins are about:

(a) 16 mm
(b) 22 mm
(c) 28 mm
(d) 36 mm

Q152. Hitch categories are specified in:

(a) IS 6284
(b) IS 4468 / ISO 730
(c) IS 9020
(d) ISO 500

Q153. The virtual hitch point of a mounted implement is the:

(a) Drawbar pin
(b) Convergence point of the lower links
(c) Top-link ball
(d) PTO end

Q154. Best weight transfer to the tractor is given by which implement attachment?

(a) Trailed
(b) Semi-mounted
(c) Fully mounted
(d) Winched

Q155. Lateral levelling of a mounted implement is set by the:

(a) Top link
(b) Right lift-rod levelling lever
(c) Sway chain
(d) Drawbar

Q156. Implement pitch (share suction attitude) is set by the:

(a) Top-link length
(b) Tyre pressure
(c) Seat
(d) PTO

Q157. In POSITION control, feedback to the valve comes from:

(a) Draft spring
(b) Linkage/lift-arm position
(c) Wheel slip
(d) Fuel rack

Q158. In DRAFT control the quantity held constant is:

(a) Depth (b) Height

(c) Implement draft (pull) (d) Speed

Q159. Draft is sensed on small tractors by:

(a) Radar (b) Top-link compression spring

(c) Ammeter (d) Airflow

Q160. On modern/heavier tractors draft sensing is via:

(a) Lower-link load pins (electronic EHC) (b) Rope

(c) Vacuum (d) Belt

Q161. Under draft control in a hard soil patch, the implement is automatically:

(a) Deepened (b) Raised slightly

(c) Stopped (d) Widened

Q162. A key traction benefit of draft control is:

(a) Less fuel tank use (b) Automatic weight transfer to rear wheels reducing slip

(c) Wider tyres (d) Cooler engine

Q163. Rotavator working depth is normally managed with:

(a) Draft control (b) Position control (+ depth/skid wheels)

(c) No control (d) Brakes

Q164. MIXED control blends:

(a) Oil and water (b) Position and draft signals

(c) Petrol and diesel (d) Front and rear brakes

Q165. Hydraulic lift capacity is rated at the hitch points and at:

(a) 310 mm (b) 610 mm behind them

(c) 1 m ahead (d) The seat

Q166. The correct-steering (Ackermann) condition is:

(a) $\cot \delta o + \cot \delta i = B/L$ (b) $\cot \delta o - \cot \delta i$ = track/wheelbase

(c) $\tan \delta o = \tan \delta i$ (d) $\delta o = \delta i$

Q167. On a turn, the INNER front wheel is steered:

(a) Less than outer (b) More than outer

(c) Same (d) Not at all

Q168. Common tractor steering gears are:

(a) Rack only (b) Worm & sector / recirculating ball

(c) Chain (d) Bevel box

Q169. Toe-in of tractor front wheels is about:

(a) 0 (b) 3–12 mm

(c) 30–50 mm (d) 100 mm

Q170. Outward tilt of the wheel top from vertical is:

(a) Caster (b) Camber

(c) Toe-out (d) KPI

Q171. Self-centring of steering after a turn is chiefly due to:

(a) Camber (b) Caster (backward kingpin tilt)

(c) Toe-in (d) Wide tyres

Q172. Full hydrostatic power steering uses:

(a) A drag link to wheels
(b) A metering (orbital) valve and steering cylinder with NO mechanical drive to wheels
(c) Chains
(d) Cable

Q173. Power steering is especially valuable with:

(a) Empty trailers
(b) Front-end loaders and puddled-field work
(c) Road transport only
(d) Small implements

Q174. Over-sensitive draft response causes:

(a) Smooth work
(b) Hunting (rhythmic raise-lower) of the linkage
(c) Stall
(d) Slip

Q175. If the front of the tractor begins to lift while pulling, the operator must FIRST:

(a) Accelerate
(b) Disengage the clutch immediately
(c) Raise the implement
(d) Turn sharply

Answers

Q141 — (b) W·x/L

Q142 — (c) 12 kN — *R_rear = W(L − x)/L = 20 × 1.2/2 = 12 kN.*

Q143 — (c) 35:65

Q144 — (b) P·h/L

Q145 — (b) 1.2 — *ΔR = 6 × 0.4/2 = 1.2 kN.*

Q146 — (b) Hitching high above the drawbar and rear-axle torque reaction — *A pull line above the rear-axle centre (or wheels trapped in mud) creates a moment that rotates the chassis about the rear axle.*

Q147 — (b) 20%

Q148 — (b) Useful for rear traction but dangerous in excess

Q149 — (b) Two lower draft links + one top link

Q150 — (b) 460 mm

Q151 — (c) 28 mm

Q152 — (b) IS 4468 / ISO 730

Q153 — (b) Convergence point of the lower links

Q154 — (c) Fully mounted

Q155 — (b) Right lift-rod levelling lever

Q156 — (a) Top-link length

Q157 — (b) Linkage/lift-arm position — *Position control = geometry feedback, constant HEIGHT.*

Q158 — (c) Implement draft (pull) — *Draft control = force feedback, constant PULL (auto depth + weight transfer).*

Q159 — (b) Top-link compression spring

Q160 — (a) Lower-link load pins (electronic EHC)

Q161 — (b) Raised slightly

Q162 — (b) Automatic weight transfer to rear wheels reducing slip

Q163 — (b) Position control (+ depth/skid wheels)

Q164 — (b) Position and draft signals

Q165 — (b) 610 mm behind them

Q166 — (b) cot δo − cot δi = track/wheelbase — *Geometry forces the inner wheel through a larger angle so both front wheels aim at the common centre on the rear-axle line.*

Q167 — (b) More than outer

Q168 — (b) Worm & sector / recirculating ball

Q169 — (b) 3–12 mm

Q170 — (b) Camber

Q171 — (b) Caster (backward kingpin tilt)
Q172 — (b) A metering (orbital) valve and steering cylinder with NO mechanical drive to wheels
Q173 — (b) Front-end loaders and puddled-field work
Q174 — (b) Hunting (rhythmic raise-lower) of the linkage
Q175 — (b) Disengage the clutch immediately — *The clutch is the instant remedy — raising the implement takes longer.*

Tractor tests, emissions, recent trends, operation and maintenance (Q176–Q210)

Q176. The most useful single performance criterion for farm tractors is:
(a) Engine rpm
(b) Maximum drawbar power
(c) Paint quality
(d) Mass

Q177. For a tractor mainly running PTO implements, the criterion of most interest is:
(a) Drawbar pull
(b) Maximum PTO power
(c) Turning circle
(d) Ground clearance

Q178. The Nebraska tractor test law was enacted in:
(a) 1905
(b) 1919
(c) 1947
(d) 1986

Q179. Nebraska tests are conducted by:
(a) USDA
(b) University of Nebraska
(c) SAE
(d) John Deere

Q180. Nebraska began reporting operator-ear sound level in dB(A) from:
(a) 1955
(b) 1971
(c) 1986
(d) 2000

Q181. In official tests, maximum drawbar pull is reported at a slip limit of:
(a) 5%
(b) 10%
(c) 15%
(d) 25%

Q182. The tractor tested at Nebraska is selected by:
(a) Random pick from dealers
(b) The manufacturer
(c) Farmers
(d) Lottery

Q183. That selection policy is justified because it:
(a) Hides defects
(b) Reduces assembly-line variability, making model comparisons exact
(c) Saves money
(d) Speeds testing

Q184. The 1986 Nebraska revision allows testing at:
(a) Any garage
(b) An OECD test station per OECD or SAE J708/ASAE S209 codes
(c) Farms
(d) Dealer sheds

Q185. A key public feature of Nebraska/OECD testing is:
(a) Secret results
(b) Published test results
(c) No records
(d) Verbal reports

Q186. Internationally accepted tractor test codes are issued by:

(a) WTO
(b) OECD
(c) IMF
(d) FAO

Q187. The famous UK station that carried out OECD/EEC/ISO tractor tests is at:

(a) Oxford
(b) Silsoe (NIAE)
(c) London
(d) Leeds

Q188. Measured tractor power is corrected for:

(a) Paint colour
(b) Atmospheric temperature and pressure to standard conditions
(c) Operator weight
(d) Tyre brand

Q189. Constant-power (torque-rise) claims are verified preferably at the:

(a) Drawbar
(b) PTO across the claimed speed range
(c) Fuel tank
(d) Radiator

Q190. India's central tractor testing institute is at:

(a) Hisar
(b) Budni (MP)
(c) Anantapur
(d) Bhopal

Q191. CFMTTI Budni was established in:

(a) 1947
(b) 1955
(c) 1971
(d) 1986

Q192. A favourable Budni test report matters commercially because it:

(a) Is decorative
(b) Underpins subsidy/finance eligibility of the model
(c) Sets fuel price
(d) Fixes tax

Q193. The two chief diesel pollutants in tension with each other are:

(a) CO and HC
(b) NOx and PM
(c) O_2 and N_2
(d) SO_2 and CO_2

Q194. Thermal NOx forms mainly due to:

(a) Low temperature
(b) High combustion temperature with excess O_2
(c) Rich mixtures
(d) Cold starts

Q195. EGR reduces NOx by:

(a) Adding oxygen
(b) Lowering peak combustion temperature with recirculated exhaust
(c) Filtering
(d) Urea

Q196. Diesel particulate matter is captured by the:

(a) DOC
(b) DPF
(c) EGR
(d) Muffler

Q197. SCR systems reduce NOx using:

(a) Petrol
(b) Urea solution (DEF/AdBlue)
(c) Water
(d) Oil

Q198. The DOC primarily oxidises:

(a) NOx
(b) HC and CO
(c) N_2
(d) Soot only

Q199. TREM-IV norms (Jan 2023) applied to tractors:

(a) Of all sizes
(b) Above 50 hp
(c) Below 20 hp
(d) Only combines

Q200. TREM-V starts for 56–560 kW tractors and combines from:

(a) Apr 2020
(b) Jan 2023
(c) 1 Oct 2026
(d) 2032

Q201. Under TREM-V the PM limit for major categories becomes:

(a) 0.15
(b) 0.025
(c) 0.015
(d) 0.005 g/kWh

Q202. India's indigenous CSIR e-tractor is the:

(a) PRIMA ET11
(b) Bharat 100
(c) Kisan-e
(d) Vidyut 5

Q203. In dusty Indian conditions, the single most critical daily maintenance item is:

(a) Seat cushion
(b) Air cleaner service
(c) Horn
(d) Mirror

Q204. Engine oil and filter are typically changed every:

(a) 10 h
(b) 50 h
(c) 250 h
(d) 2000 h

Q205. Transmission-hydraulic oil and filters are typically changed at:

(a) 100 h
(b) 250 h
(c) 500 h
(d) 1000 h

Q206. Fan-belt tension and greasing of points are done about every:

(a) 10 h
(b) 50 h
(c) 500 h
(d) Never

Q207. Power tillers commonly fall in the power range:

(a) 2–4 hp
(b) 7–12 hp
(c) 20–30 hp
(d) 40–50 hp

Q208. The principal working attachment of a power tiller is the:

(a) MB plough
(b) Rotary tiller
(c) Baler
(d) Sprayer

Q209. For long off-season storage, the battery should be:

(a) Left connected
(b) Removed and charged periodically
(c) Drained of acid and left dry forever
(d) Frozen

Q210. Dust ingestion past a faulty air cleaner causes:

(a) Better cooling
(b) Rapid abrasive wear of liners and rings ('dusting')
(c) Higher octane
(d) Cleaner oil

Answers

Q176 — (b) Maximum drawbar power — *Liljedahl p. 403.*

Q177 — (b) Maximum PTO power — *Liljedahl p. 403.*

Q178 — (b) 1919 — *Liljedahl p. 420 — the 1919 law.*

Q179 — (b) University of Nebraska

Q180 — (b) 1971 — *Liljedahl p. 404 — 1971, dB(A) at the operator's ear.*

Q181 — (c) 15% — *Liljedahl p. 431 — maximum pull at the 15% slip limit.*

Q182 — (b) The manufacturer

Q183 — (b) Reduces assembly-line variability, making model comparisons exact — *Liljedahl pp. 431–432, Fig. 14-16 — random same-model tractors spread ~±5% in power.*

Q184 — (b) An OECD test station per OECD or SAE J708/ASAE S209 codes — *Liljedahl p. 432 — 1986 LB 708 revision (OECD stations, SAE J708/ASAE S209, published results, parts availability).*

Q185 — (b) Published test results

Q186 — (b) OECD

Q187 — (b) Silsoe (NIAE)

Q188 — (b) Atmospheric temperature and pressure to standard conditions — *Liljedahl p. 433 — atmospheric correction.*

Q189 — (b) PTO across the claimed speed range — *Liljedahl p. 431 — constant-power claims verified at the PTO.*

Q190 — (b) Budni (MP)

Q191 — (b) 1955

Q192 — (b) Underpins subsidy/finance eligibility of the model

Q193 — (b) NOx and PM

Q194 — (b) High combustion temperature with excess O_2

Q195 — (b) Lowering peak combustion temperature with recirculated exhaust

Q196 — (b) DPF

Q197 — (b) Urea solution (DEF/AdBlue)

Q198 — (b) HC and CO

Q199 — (b) Above 50 hp

Q200 — (c) 1 Oct 2026 — *TREM dates as given in the source notes (MoRTH notifications, verified July 2026).*

Q201 — (c) 0.015

Q202 — (a) PRIMA ET11

Q203 — (b) Air cleaner service

Q204 — (c) 250 h

Q205 — (d) 1000 h

Q206 — (b) 50 h

Q207 — (b) 7–12 hp

Q208 — (b) Rotary tiller

Q209 — (b) Removed and charged periodically

Q210 — (b) Rapid abrasive wear of liners and rings ('dusting')